CauSAIL: Causal Graph-Grounded Multimodal LLMs for Maritime Supply Chain Disruption Detection

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ARTICLE INFO

Keywords:
Maritime Disruption Detection
Multimodal LLMs
Knowledge Graph
RAFT
LLM Bias-Aware Fusion

ABSTRACT

Maritime supply chain disruptions arise from complex interactions among vessel movements, port operations, satellite observations, public events, trade dependencies, corporate filings, and global pressure indicators. Existing monitoring systems often collapse these diverse modalities into static feature representations and rely on Retrieval-Augmented language models without sufficient causal grounding, modality control, and feature verification. We present CauSAIL that addresses these gaps through four connected modules: (1) integration of seven open-access datasets through a port-month alignment pipeline; (2) develop a Causal Temporal Supply Chain Knowledge Graph (CT-SCKG) and build Retrieval-Augmented Fine-Tuning (RAFT)-based feature sets to ground LLM decisions in structured and textual features; (3) develop a Suppression-Focal Modality-Aware LAFAM module (MA-LAFAM) to combine diverse modalities by handling missing modalities and class imbalance; (4) propose mathematical bias-mitigation methods to reduce seven IBM-recognised LLM bias types. Experimental results show that CauSAIL effectively identifies port-month disruption patterns by connecting vessel traces, satellite imagery, supply-chain relations, and retrieved documents. Ablation study demonstrates that vessel movement Automatic Identification System (AIS) features provide major disruption signal, and CT-SCKG, RAFT feature, MA-LAFAM, and mathematical bias-mitigation methods improve robustness and reduce unsupported LLM decisions. These experimental results suggest that reliable maritime intelligence systems benefit from grounded system design rather than large language model scale alone. We release our data and code here [1].

1. Introduction

Maritime supply chains are dynamic industrial systems. A disruption can arise from port congestion, vessel slowdowns, trade shocks, extreme weather, labor events, geopolitical conflict, supplier risk, or a combination of these factors [30]. Automatic Identification System (AIS) tracks show vessel slowdown and dwell patterns [30]. Satellite observations reflect changes near port waters [4]. Event streams report strikes, conflict, sanctions, and congestion [30]. Trade records reveal dependencies among products and partner countries [37]. Company filings provide supplier and logistic risk [30]. Existing monitoring systems do not consider all the factors at the same time. Many reduce diverse features to static feature tables [20, 50, 60] and others depend on Retrieval-Augmented Language models with limited control over feature quality [26, 37, 39]. This existing workflow creates three research gaps. First, diverse features lack a shared decision across time and place. Second, the retrieved test includes irrelevant information, which raises the risk of unsupported LLMs outputs. Third, they do not address the real deployment settings because of missing modalities, severe label imbalance, and limited hardware budgets. We study these gaps and focus on five research questions.

- RQ1. How can diverse maritime supply chain feature be aligned into a single operational decision unit?
- RQ2. How can causal Knowledge Graphs (KG) and RAFT feature system reduce unsupported LLM decisions in logistics risk monitoring?
- RQ3. How can AIS vessel features, Sentinel-2 satellite features, KG embeddings, and LLM hidden states be integrated when some modalities are missing?

[1]https://github.com/mdsajeebulislamsk/CauSAIL
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[1]These authors contributed equally to this work.
[2]This research received no grant from any funding agency in the public, commercial, or not-for-profit sectors.

- RQ4. How can mathematical methods mitigate LLM bias types acknowledged by IBM?
- RQ5. Which data sources drive positive disruption detection, and which components reduce false positives?

To meet these goals, we propose Causal Supply-chain Artificial Intelligence for Logistics (CauSAIL). The main contributions of our study are as follows:

- We propose CauSAIL, a causal graph-grounded multimodal LLM framework for port-month maritime supply chain disruption detection.
- We build a port-month alignment pipeline that integrates seven open-access maritime and supply chain datasets into one operational decision unit.
- We develop a Causal Temporal Supply Chain Knowledge Graph (CT-SCKG) that connects ports, months, companies, products, trade flows, events, filings, and local observations in a time-aware structure to support grounded reasoning.
- We assemble RAFT-based feature sets with core and tangent documents that improve feature selection and reduce reliance on irrelevant text.
- We develop a Suppression-Focal Modality-Aware Fusion Module (MA-LAFAM) to combine AIS features, Sentinel-2 signals, KG embeddings, and LLM hidden states under missing modality and class imbalance settings.
- We propose mathematical bias-mitigation methods to reduce seven types of LLM bias and show that they improve robustness and reduce unsupported decisions.

2. Related Work

Prior work on maritime disruption detection was developed on individual modalities such as AIS movement [57], satellite imagery [11], public events [22], trade records [61], company filings [49], and retrieved text [29].

2.1. Retrieval-Augmented Fine-Tuning (RAFT) Based Language Models

Retrieval-Augmented Generation (RAG) connects a parametric language model with an external document collection. Lewis et al. show that retrieval improves knowledge-intensive generation by giving the model access to non-parametric memory [36]. This idea fits supply-chain intelligence because relevant evidence changes across months, ports, firms, trade flows, and public events. RAG performance depends on retrieval quality. In maritime monitoring, a port-month query often retrieves documents from a different port, an old event period, a broad trade report, or a weakly related filing. This creates a risk that the LLM will rely on irrelevant or stale context. Recent supply-chain studies use retrieval-augmented LLMs to connect external evidence with risk analysis, but retrieval noise remains a practical concern in operational settings [26, 37, 39].

Retrieval-Augmented Fine-Tuning (RAFT) addresses this problem by training the model with relevant documents and distractor documents [63]. The relevant documents teach the model which evidence supports the answer. The distractor documents help model to ignore weak or unrelated context. CauSAIL follows this direction at the port-month level. It builds core evidence documents from AIS, Sentinel-2, KG, event, trade, and filing signals, and it builds tangent documents from weaker evidence. This design supports one practical goal: the LLM should learn which retrieved documents support the disruption decision, rather than trusting every retrieved passage.

2.2. Knowledge graph Grounding for LLM Decisions

Knowledge graphs (KGs) represent entities and relations as connected nodes and edges. Prior surveys show that KGs support entity linking, relation modeling, semantic search, and reasoning over structured knowledge [21]. Recent KG-LLM studies further show that graph structure helps language models ground their outputs in external knowledge instead of relying only on parametric memory [45]. Prior supply chain studies use graphs to model firms, products, suppliers, and trade dependencies [8]. These works show the value of relational structure for risk analysis. We use this method to link between vessels, ports, months, countries, companies, products, trade flows, filings, and events. A flat feature table stores these features in isolated columns, but it does not preserve the relationships among them, and a text only LLM is also unstable to link among these entities. So, we develop a Causal Temporal Supply Chain Knowledge Graph (CT-SCKG) to connect Port, Month, Country, Company, Product, Tradeflow, Event, Filing, Disruption Month, and Local Observation nodes. We use causal structure as domain grounding.

2.3. Multimodal Fusion for Maritime and Event Intelligence

Prior maritime studies use AIS traces to estimate vessel traffic, berth stay, congestion, port efficiency, and movement anomalies [41, 50]. Satellite-based studies use optical or radar imagery to detect ships, observe port activity, and estimate maritime trade signals from space [31, 54, 55]. Event detection studies combine public reports, news streams, and structured event databases to identify conflict, sanctions, strikes, disasters, and logistics shocks [24, 34, 35]. These studies show the value of physical and textual evidence, but they work on each source separately. In CauSAIL, we address this gap through a modality-aware fusion design. We study AIS-derived vessel features, Sentinel-2 satellite indicators, CT-SCKG embeddings, and RAFT-guided LLM hidden states for a single industrial prediction task.

2.4. Bias-Aware Objectives and Reliability

LLM-based decision systems face bias from data, representation, retrieval, labels, and deployment context. IBM's AI Fairness 360 work provides a framework for measuring and mitigating unwanted algorithmic bias in industrial settings [5]. Recent LLM studies also report bias in generated content, evaluation behaviour, and decision support [33, 56]. We address these risks through mathematical bias-mitigation methods. Orthogonal projection reduces direct reliance on port, time, or modality identity [3]. MMD regularization reduces group-level representation shift [18]. Evidence alignment links LLM hidden states with KG evidence. Counterfactual consistency reduces unstable predictions under evidence changes. Calibration loss reduces overconfident outputs [19]. Suppression-Focal loss focuses learning on hard disruption cases and class imbalance [38]. Fourth-Order Runge-Kutta (RK4) method hidden-state refinement stabilizes hidden representations before classification, following the use of differential equation refinement in neural models [9].

2.5. Open LLM Backbones and Efficient Fine-Tuning

Open LLMs have made industrial NLP systems easier to deploy under local hardware and privacy constraints. Qwen2.5 provides instruction-tuned models across several sizes and reports strong performance in instruction following, structural data analysis, reasoning, and long-text generation [62]. The Llama model provides open-weight language models for general language understanding and reasoning tasks [13]. Gemma 2 studies open models at practical sizes and focuses on efficient performance for downstream use [17]. Phi-3 studies compact language models for local and resource-constrained deployment [1]. GPT-OSS-20B provides a larger open-weight comparison point for local inference and private deployment [7]. DeepSeek-VL provides an open vision-language model for real-world image-text understanding [40]. GPT-2 remains a standard Transformer baseline for generative pretraining [47]. Efficient adaptation methods reduce the cost of training these models. LoRA freezes the pretrained weights and trains low-rank adapter matrices, thereby reducing trainable parameters and memory usage [23]. QLoRA [53] extends this idea with 4-bit quantization, NF4 weights, double quantization, and paged optimizers, which makes large-model fine-tuning possible on smaller GPUs [12]. Hyperparameter search also matters in low-resource fine-tuning. Optuna provides a define-by-run optimization framework with pruning and sampling strategies for efficient hyperparameter selection [2].

3. Methodology

We present the full workflow of CauSAIL in Figure 1. This figure presents the complete path from data preprocessing to final port-month disruption prediction.

3.1. Data Preprocessing, Feature Extraction, and Port-Month Alignment

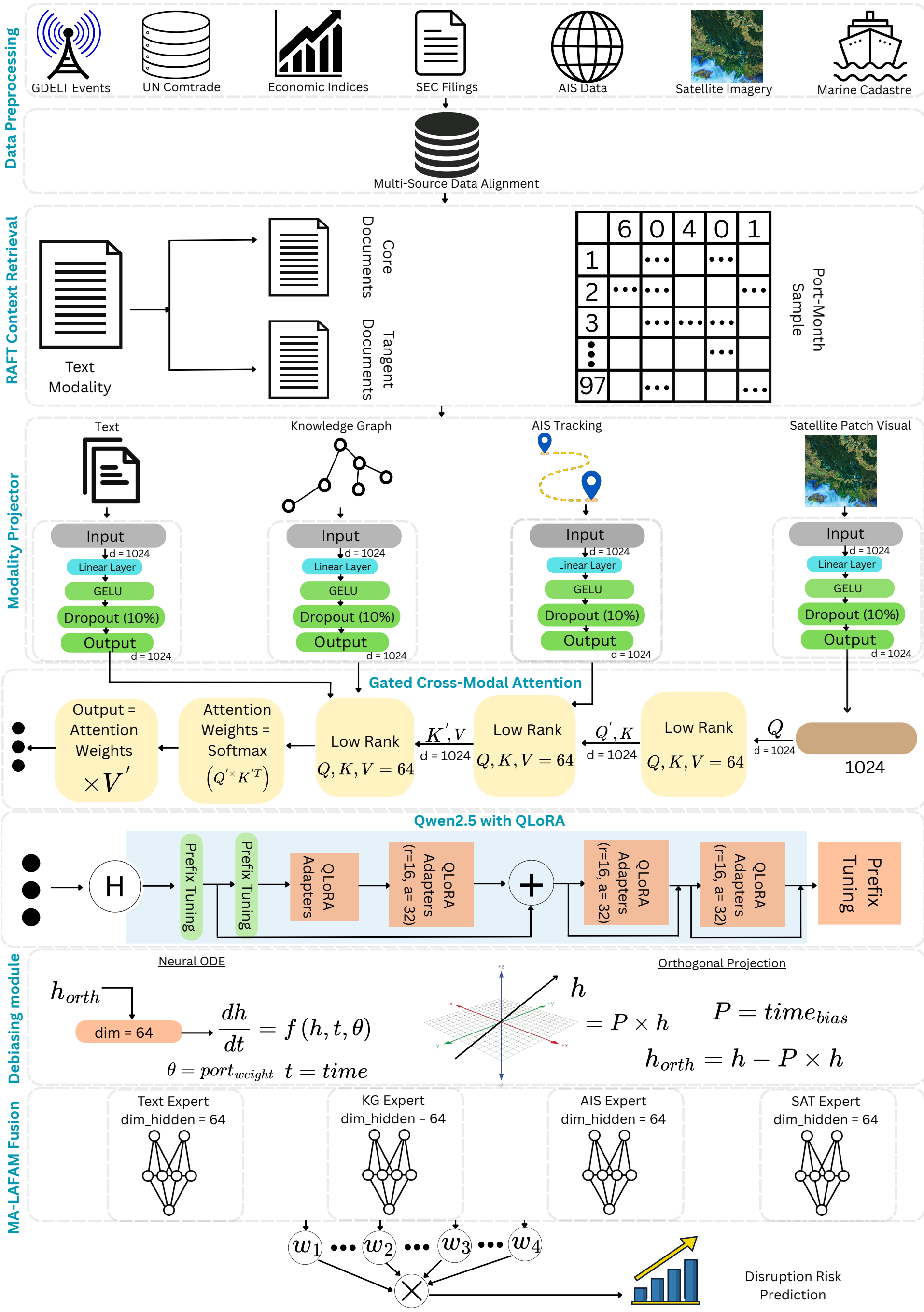
We consider each port in each month as a single prediction sample. Let $p \in \mathcal{P}$ be a normalized port key and let $t \in \mathcal{T}$ be a calendar month. We represent each sample as

$$x_{p,t} = \{X^{AIS}_{p,t}, X^{SAT}_{p,t}, X^{GSCPI}_{t}, X^{EVT}_{p,t}, X^{TRD}_{p,t}, X^{FIL}_{p,t}, X^{KG}_{p,t}, X^{TXT}_{p,t}\} \tag{1}$$

where $X^{AIS}_{p,t}$, $X^{SAT}_{p,t}$, X^{GSCPI}_{t}, $X^{EVT}_{p,t}$, $X^{TRD}_{p,t}$, $X^{FIL}_{p,t}$, $X^{KG}_{p,t}$, and $X^{TXT}_{p,t}$ denote AIS, Sentinel-2, global pressure, public event, trade, filing, graph, and retrieved-text evidence, respectively. The label is

$$y_{p,t} \in \{0, 1\} \tag{2}$$

Figure 1: CauSAIL workflow for port-month maritime supply chain disruption detection.



where $y_{p,t} = 1$ denotes a disrupted port-month and $y_{p,t} = 0$ denotes a normal port-month.

By following this process, we match the temporal nature of maritime disruption. A port can be normal in one month and risky in another. The equation 1 also allows for lagged variables, chronological splits, and evidence selection within the same time window. We represent the seven datasets and their modalities in Table 1.

Before alignment, we extract modality-specific variables. AIS provides vessel observation count, unique vessel count, active days, mean speed, low-speed ratio, stationary ratio, and heading entropy. Sentinel-2 provides NDWI, NDVI, water fraction, vessel proxy, brightness, NIR, cloud cover, and valid pixel rate. GSCPI, GDELT, Comtrade, and

Table 1
Seven datasets integrated into CauSAIL. Each dataset is mapped to the shared port-month key $k = (p, t)$.

No.	Dataset	Modality	Source
1	GDELT Events 2.0	Event and text modality: conflict, trade, geopolitical events, sanctions, strikes, congestion, source URLs, and event tone.	[35]
2	UN Comtrade	Trade modality: bilateral trade flows, product dependency, partner-country exposure, and commodity-level risk signals.	[58]
3	SEC EDGAR	Filing and text modality: 10-K and 8-K supply-chain risk mentions, supplier risk, production risk, and logistics risk.	[59]
4	NY Fed GSCPI	Economic-index modality: monthly global supply-chain pressure, z-score, lagged pressure, and rolling pressure context.	[6]
5	U.S. DOT and NOAA AIS	Vessel-movement modality: dwell behaviour, speed, heading entropy, and local maritime anomaly signals.	[43]
6	Sentinel-2 L2A optical imagery	Satellite modality: NDWI, NDVI, water fraction, vessel proxy, brightness, NIR, cloud cover, and valid pixel features.	[14, 15]
7	NOAA MarineCadastre AIS	AIS modality: monthly vessel counts, active days, mean speed, low-speed ratio, stationary ratio, and heading entropy.	[44]

SEC EDGAR provide pressure indicators, event signals, trade-dependency evidence, and company risk text for the construction of CT-SCKG and RAFT. We normalize port names to reduce duplicate identifiers. For example, New York/NJ maps to `new york` and `new jersey`, Norfolk maps to `norfolk hampton roads`, and Portland OR maps to `portland oregon`. We keep Los Angeles and Long Beach separate because their AIS and satellite patterns differ. We then map all records to the monthly level. AIS timestamps are aggregated into monthly vessel variables. Sentinel-2 acquisitions are assigned to their corresponding month. GSCPI is reported at the monthly level. GDELT events are grouped by event month. SEC filings use the filing month. Comtrade records are linked through country, product, and reporting period. Each record r_i is mapped by

$$\phi(r_i) \rightarrow (p, t) \tag{3}$$

where p is the normalized port key and t is the month. The final aligned dataset is

$$\mathcal{D} = \{(x_{p,t}, y_{p,t}) \mid p \in \mathcal{P}, t \in \mathcal{T}\} \tag{4}$$

where $\mathcal{D}$ denotes the full dataset, $x_{p,t}$ denotes the aligned representation for port p in month t, and $y_{p,t}$ denotes the disruption label. The sets $\mathcal{P}$ and $\mathcal{T}$ denote all normalized ports and month.

3.2. Feature Extraction

For AIS, we aggregate vessel records by $k = (p, t)$. We compute observation count, unique vessel count, active days, mean speed over ground, low-speed ratio, stationary ratio, and heading entropy. The low-speed ratio is the fraction of AIS records below a speed threshold, and the stationary ratio is the fraction of records with near-zero movement.

For Sentinel-2, we summarize optical imagery over the port region. We compute NDWI and NDVI from spectral bands, then derive the water fraction, a vessel proxy, brightness statistics, NIR mean, cloud cover, and the valid pixel rate. These variables represent port-water and visible surface conditions.

For GSCPI, we use the current monthly value, z-score, absolute z-score, lag values, and rolling statistics. For GDELT, we keep disruption-relevant event records and summarize event pressure, mentions, tone, and event categories. For Comtrade, we create country-product trade dependency features. For SEC EDGAR, we extract filing text related to supplier, production, logistics, and supply chain risk.

3.3. CT-SCKG and RAFT Evidence Construction

After alignment, we develop a Causal Temporal Supply Chain Knowledge Graph (CT-SCKG) to capture relations among ports, months, events, companies, products, trade flows, filings, and local observations. We define the graph as

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \tau) \tag{5}$$

where $\mathcal{V}$ is the node set, $\mathcal{E}$ is the edge set, $\mathcal{R}$ is the relation set, and τ stores the temporal context of each edge. Each edge is represented as

$$e_i = (v_s, r_i, v_o, t_i) \tag{6}$$

where v_s and v_o are source and target nodes, r_i is the relation type, and t_i is the associated month. For each port-month sample (p, t), we retrieve the linked graph evidence $\mathcal{G}_{p,t}$ and encode it as a KG representation $h_{p,t}^{KG}$. This representation provides the model structured context about which events, filings, trade flows, products, and local observations are connected to the target port-month.

We also construct RAFT-based evidence for the LLM branch. For each query $q_{p,t}$, we build core documents $\mathcal{C}_{p,t}$ from relevant evidence and tangent documents $\mathcal{T}_{p,t}$ from weaker or distracting evidence. The RAFT input is

$$R_{p,t} = [q_{p,t}; \mathcal{C}_{p,t}; \mathcal{T}_{p,t}] \tag{7}$$

and the LLM produces the textual representation

$$h_{p,t}^{LLM} = f_{LLM}(R_{p,t}) \tag{8}$$

The equations 7 and 8 improve the model connecting structured supply-chain relations with retrieved textual evidence while reducing unsupported reliance on irrelevant documents [63].

3.4. Modality Projection and LLM Encoding

After constructing CT-SCKG and RAFT evidence, we map all evidence sources into a shared hidden space. We use four main branches: AIS, satellite, KG, and RAFT-guided text. For each port-month sample (p, t), the model receives

$$\{X_{p,t}^{AIS}, X_{p,t}^{SAT}, h_{p,t}^{KG}, R_{p,t}\} \tag{9}$$

where $X_{p,t}^{AIS}$ and $X_{p,t}^{SAT}$ are structured physical features, $h_{p,t}^{KG}$ is the graph representation, and $R_{p,t}$ is the RAFT evidence input. Each non-text modality is projected into the LLM hidden dimension. We define

$$h_{p,t}^{m} = \text{Dropout}(\text{GELU}(W_m X_{p,t}^{m} + b_m)) \tag{10}$$

where $m \in \{AIS, SAT, KG\}$, W_m and b_m are modality-specific projection parameters, and $h_{p,t}^{m}$ is the projected hidden state. The RAFT evidence is encoded by the LLM as

$$h_{p,t}^{LLM} = f_{LLM}(R_{p,t}) \tag{11}$$

We use Qwen2.5 as the main language backbone and fine-tune it with QLoRA. The projected AIS, satellite, and KG states are then combined with the LLM hidden state:

$$H_{p,t} = \{h_{p,t}^{AIS}, h_{p,t}^{SAT}, h_{p,t}^{KG}, h_{p,t}^{LLM}\} \tag{12}$$

We apply gated cross-modal attention over $H_{p,t}$. It allows textual evidence to interact with physical and graphical evidence before final fusion. Now, the model connects vessel slowdown, satellite context, KG relations, and RAFT documents within the same port-month sample.

Algorithm 1 MA-LAFAM fusion

Input: $h^{AIS}, h^{SAT}, h^{KG}, h^{LLM}$ mask a
Output: disruption probability $\hat{y}$
$z^m \leftarrow E_m(h^m)$ for each modality m
$g \leftarrow softmax(W_g[h^{AIS}; h^{SAT}; h^{KG}; h^{LLM}])$
$\tilde{g} \leftarrow Normalize(a \odot g)$
$h^* \leftarrow \sum_m \tilde{g}^m z^m$
$\hat{y} \leftarrow \sigma(W_o h^* + b_o)$
return $\hat{y}$

3.5. MA-LAFAM Fusion and Prediction

LaFAM is introduced as a label-free activation-map method for feature attribution in CNNs [32]. We develop Suppression-Focal Modality-Aware LAFAM (MA-LAFAM) for port-month disruption prediction. MA-LAFAM combines modality experts, masking, gated fusion, and Suppression-Focal learning to handle missing evidence and class imbalance. For each port-month sample (p, t), MA-LAFAM receives four hidden states:

$$H_{p,t} = \{h_{p,t}^{AIS}, h_{p,t}^{SAT}, h_{p,t}^{KG}, h_{p,t}^{LLM}\} \tag{13}$$

Each modality is first passed through a separate expert:

$$z_{p,t}^{m} = E_m(h_{p,t}^{m}); m \in \{AIS, SAT, KG, LLM\} \tag{14}$$

A gate network estimates the contribution of each modality:

$$g_{p,t} = softmax(W_g[h_{p,t}^{AIS}; h_{p,t}^{SAT}; h_{p,t}^{KG}; h_{p,t}^{LLM}]) \tag{15}$$

We use a mask $a_{p,t} \in \{0, 1\}^4$ to suppress missing evidence. The masked gate is

$$\tilde{g}_{p,t}^{j} = \frac{a_{p,t}^{j} g_{p,t}^{j}}{\sum_{\ell=1}^{4} a_{p,t}^{\ell} g_{p,t}^{\ell} + \epsilon} \tag{16}$$

The fused representation and prediction are

$$h_{p,t}^{*} = \sum_{j=1}^{4} \tilde{g}_{p,t}^{j} z_{p,t}^{j} \tag{17}$$

$$\hat{y}_{p,t} = \sigma(W_o h_{p,t}^{*} + b_o) \tag{18}$$

We train the module with Suppression-Focal loss:

$$\mathcal{L}_{SF} = -\alpha_y S(1 - p_y)^{\gamma} \log(p_y) \tag{19}$$

where p_y is the predicted probability of the true class, α_y weights rare disruption cases, γ focuses learning on difficult samples, and S suppresses majority-class samples. This process allows MA-LAFAM to weight the available modalities for each port-month and reduce the effect of weak evidence. We represent the fusion process in Algorithm 1.

Table 2
Mapping between IBM-recognized LLM bias types and CauSAIL mitigation methods, following IBM bias categories and fairness guidance [5, 25, 27].

Bias type	Risk in CauSAIL	Mitigation method
Data bias	Some ports or months have richer AIS, satellite, or text evidence.	Modality masks, MMD regularization, and missing-modality checks [18].
Selection bias	Training samples may overrepresent specific ports or disruption periods.	Chronological split design and group-level regularization [18].
Representation bias	Hidden states may encode port, month, or modality identity shortcuts.	Orthogonal projection over port, time, and modality bias bases [18].
Confirmation bias	Retrieved text may reinforce weak or incorrect evidence of disruption.	RAFT tangent documents and KG-LLM evidence alignment [63].
Automation bias	The model may produce high-confidence unsupported alerts.	Calibration loss and evidence-linked prediction output [19].
Evaluation bias	A single split or metric may overstate reliability.	Ablation tests, bootstrap confidence intervals, and seed checks [46].
Deployment bias	Missing AIS, cloudy satellite scenes, or incomplete KG evidence may change behavior.	Modality masks and counterfactual consistency [9].

3.6. Bias-Aware Hidden-State Regularization

We add bias-aware regularization to reduce shortcut dependence during training. In equation 20, $h_{p,t}$ denote the fused hidden representation for port p in month t. We first remove the component of $h_{p,t}$ that is aligned with a learned bias direction b:

$$h^{\perp}_{p,t} = h_{p,t} - \mathrm{proj}_b(h_{p,t}) \tag{20}$$

The equation 20 is used to reduce direct dependence on identifiers such as port, time, or modality availability. We then train the model with the following objective:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{SF} + \lambda_{mmd}\mathcal{L}_{MMD} + \lambda_{evid}\mathcal{L}_{evid} + \lambda_{cf}\mathcal{L}_{cf} \\ + \lambda_{cal}\mathcal{L}_{cal} + 0.25\mathcal{L}_{expert} + 0.10\mathcal{L}_{gate}\end{aligned} \tag{21}$$

Here, $\mathcal{L}_{SF}$ handles class imbalance, $\mathcal{L}_{MMD}$ reduces distribution differences across groups, $\mathcal{L}_{evid}$ encourages agreement between LLM and KG representations, $\mathcal{L}_{cf}$ penalizes unstable predictions under small evidence changes, and $\mathcal{L}_{cal}$ reduces overconfident probabilities. The expert and gate losses support modality-specific prediction and stable fusion. We provide details in Section 3.8.

3.7. Reliability Checks

We evaluate reliability after training to test whether CauSAIL remains stable under uncertainty in evidence and model configuration. We use bootstrap confidence intervals, Brier score calibration, seed sensitivity, port-identity shortcut testing, time-leakage auditing, and tangent-document stress testing for evaluation. The detailed results are reported in Section 5.6.

3.8. Bias-Aware Regularization Details

We map seven IBM-recognized LLM bias types to mathematical mitigation methods used in CauSAIL 2. The goal is to reduce shortcut learning, unstable evidence use, overconfidence, and class imbalance in port-month disruption prediction. The RAFT-based evidence terms are trained on relevant and distractor documents [63]. The regularization terms build on MMD [18], calibration [19], focal loss [38], and neural ODE style refinement [9].

Orthogonal projection: Let h be a hidden representation and b be a bias basis learned from port, time, or modality indicators. We remove the component of h aligned with b:

$$\mathrm{proj}_b(h) = \frac{h^\top b}{b^\top b + \epsilon} b, \tag{22}$$

$$h_\perp = h - \mathrm{proj}_b(h). \tag{23}$$

The $h_\perp$ reduces direct reliance on shortcut variables.

RK4 hidden-state refinement: We refine hidden states with a learned neural dynamics function f_θ, following neural ODE representation refinement [9, 28]. The fourth-order Runge-Kutta update is

$$\begin{aligned} k_1 &= f_\theta(h_t), \\ k_2 &= f_\theta(h_t + \frac{dt}{2} k_1), \\ k_3 &= f_\theta(h_t + \frac{dt}{2} k_2), \\ k_4 &= f_\theta(h_t + dt k_3), \end{aligned} \tag{24}$$

$$h_{t+1} = h_t + \frac{dt}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{25}$$

We use this step as representation refinement before classification.

MMD regularization: To reduce group-level representation shift, we apply Maximum Mean Discrepancy between two groups a and b [18]:

$$\mathcal{L}_{MMD} = \left\| \mu(h_a) - \mu(h_b) \right\|_2^2. \tag{26}$$

The groups can correspond to port partitions, time partitions, or modality-availability partitions.

Evidence alignment: We align LLM evidence with KG evidence using cosine distance:

$$\mathcal{L}_{evid} = 1 - \cos(h_{p,t}^{LLM}, h_{p,t}^{KG}). \tag{27}$$

The $\mathcal{L}_{evid}$ prevents the LLM branch from making decisions unsupported by graph evidence.

Counterfactual consistency: For a perturbed port-month sample $x_{p,t}^{cf}$, we minimize prediction divergence:

$$\mathcal{L}_{cf} = D_{KL}(P(y|x_{p,t}) \| P(y|x_{p,t}^{cf})). \tag{28}$$

This term reduces unstable decisions in response to small changes in evidence.

Calibration: We reduce overconfidence using a probability calibration loss, following prior work on neural network calibration [19, 48]:

$$\mathcal{L}_{cal} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2, \tag{29}$$

where $\hat{p}_i$ is the predicted disruption probability and y_i is the true label.

Table 3
Training and evaluation setup for CauSAIL.

Item	Setting	Item	Setting
Hardware	NVIDIA RTX 4070, 12 GB	Main backbone	Qwen2.5-3B-Instruct
Fine-tuning	QLoRA, 4-bit NF4, BF16	Hyperparameter search	Optuna 4.0
LoRA rank / alpha	16 / 32	Batch size	4
Learning rate	2×10^{-5}	Warmup / weight decay	0.20 / 0.02
Primary metrics	Accuracy, precision, recall	Additional metrics	F1, AUC, MCC

Suppression-Focal loss: We use Suppression-Focal loss to handle hard examples and class imbalance. This follows the focal-loss idea of increasing attention to difficult and minority-class examples [38]:

$$\mathcal{L}_{SF} = -\alpha_y S(1 - \hat{p}_y)^{\gamma} \log(\hat{p}_y), \tag{30}$$

where α_y gives higher weight to rare disruption cases, γ controls the focus on hard samples, and S suppresses easy majority-class examples.

Final objective: The full training objective is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{SF} &+ \lambda_{mmd}\mathcal{L}_{MMD} + \lambda_{evid}\mathcal{L}_{evid} \\ &+ \lambda_{cf}\mathcal{L}_{cf} + \lambda_{cal}\mathcal{L}_{cal} \\ &+ 0.25\mathcal{L}_{expert} + 0.10\mathcal{L}_{gate}.\end{aligned} \tag{31}$$

The λ terms control the strength of each regularizer. This objective combines class-imbalance handling, evidence grounding, representation stability, calibration, and modality-gate regularization.

4. Experimental Setup

We present the experimental setup of CauSAIL in Table 3. We fine-tune Qwen2.5-3B-Instruct with QLoRA [12, 62]. We use Optuna 4.0 to select hyperparameter for LLM fine-tuning [2]. We report accuracy, precision, recall, F1 [51], AUC, and MCC [46]. Metric definitions are provided in Section 4.1.

4.1. Evaluation Metric

We report standard classification metrics used for binary classification evaluation [52]. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{32}$$

$$Precision = \frac{TP}{TP + FP}. \tag{33}$$

$$Recall = \frac{TP}{TP + FN}. \tag{34}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}. \tag{35}$$

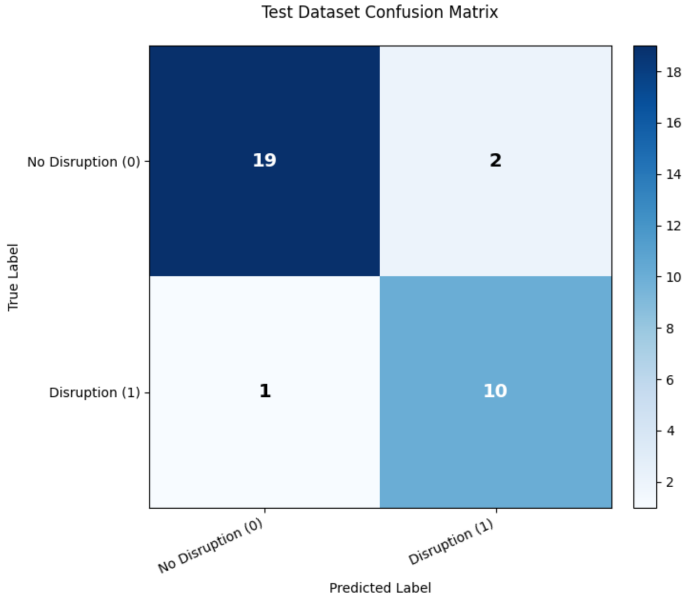


Figure 2: Confusion matrix of CauSAIL.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{\begin{aligned}&(TP + FP)(TP + FN)\\ &\cdot (TN + FP)(TN + FN)\end{aligned}}}. \tag{36}$$

AUC measures ranking quality across decision thresholds [16]. We use F1 and MCC as key metrics because disruption cases are fewer than normal cases; MCC is especially useful for imbalanced binary classification because it summarizes all four confusion-matrix terms [10, 42].

5. Results and Discussion

5.1. Main Test Performance

Table 4 reports the performance of CauSAIL on the test dataset, and it achieves 0.9062 accuracy, 0.8696 F1, 0.9610 AUC, and 0.7984 MCC. The recall score of 0.9091 indicates that CauSAIL captures most positive disruption cases. The precision of 0.8333 means that most predicted positive cases are correct. The MCC score of 0.7984 indicates a balanced classification result between disrupted and normal cases. The AUC value of 0.9610 shows that CauSAIL separates disrupted and normal port-months across decision thresholds.

Figure 2 shows the confusion matrix of CauSAIL. The model correctly identifies 19 normal port-months and 10 disrupted port-months. It misses one disruption case and raises two false disruption alerts. In an early-warning, false negatives are more costly because a missed disruption may delay response. False positives require to increase analysis of workload, but two false positives in 32 test cases are acceptable, as the system also provides KG and RAFT evidence for review.

5.2. Comparison with LLM Baselines

Table 5 compares CauSAIL performance with six LLM models. CauSAIL obtains the best accuracy, precision, F1, and AUC. Llama-3.2-3B and GPT-OSS-20B achieve perfect recall, but their lower precision shows that they produce more incorrect predictions of disruption. The comparison results show that larger LLMs do not automatically solve maritime disruption detection. GPT-OSS-20B is larger than the CauSAIL model, but it still falls behind CauSAIL in accuracy, precision, F1, and AUC. DeepSeek-VL achieves high recall but low precision and low accuracy, which means

Table 4
Main selected-local test result for CauSAIL.

Model	Accuracy	Precision	Recall	F1	AUC	MCC	TP	FP	FN	TN
CauSAIL	**0.9062**	**0.8333**	**0.9091**	**0.8696**	**0.9610**	**0.7984**	10	2	1	19

Table 5
Model performance comparison on the selected local port-month disruption detection task.

Model	Accuracy	Precision	Recall	F1	AUC
DeepSeek-VL	0.4062	0.3571	0.9091	0.5128	0.7359
GPT-OSS-20B	0.8125	0.6471	**1.0000**	0.7857	0.8918
Llama-3.2-3B	0.8438	0.6875	**1.0000**	0.8148	0.9394
Phi-3.5-mini	0.5312	0.3000	0.2727	0.2857	0.4545
Gemma-2B	0.5938	0.4500	0.8182	0.5806	0.6840
GPT-2	0.6562	0.5000	0.0909	0.1538	0.8268
CauSAIL (proposed)	**0.9062**	**0.8333**	0.9091	**0.8696**	**0.9610**

it tends to over-predict disruption. GPT-2 shows a different failure mode. Its AUC is higher than its F1, which suggests some ranking signal, but the thresholded decision is weak because recall is only 0.0909. CauSAIL improves the task by combining vessel movement evidence, satellite context, graph grounding, retrieval control, and modality-aware fusion rather than relying on language model scale alone.

5.3. Ablation Analysis

The ablation results of our study are presented in Table 6. The full model achieves the highest F1 and MCC. Removing AIS causes the largest performance collapse. In the NO_AIS setting, F1 drops from 0.8696 to 0.0000. This result shows that AIS carries the strongest positive disruption signal in the selected-local split. Vessel movement features such as dwell behaviour, low-speed ratio, stationary ratio, and heading entropy directly capture congestion port activity. From the ablation study, we get three main findings. First, AIS is necessary for positive disruption detection in this setting. AIS-only reaches 0.8000 F1, while NO_AIS, SAT_ONLY, and LLM_RAFT_ONLY fail to detect positive cases after thresholding. This means the local vessel movement signal is not only useful, but essential. Second, AIS alone is still weaker than the full model. AIS-only reaches 0.8000 F1 and 0.6880 MCC, while the full model reaches 0.8696 F1 and 0.7984 MCC. CT-SCKG links port-month states to events, filings, products, companies, and trade flows. RAFT provides controlled textual evidence. MA-LAFAM learns how much weight each signal should receive. Third, the NO_RAFT_TEXT setting shows a precision-recall trade-off. Without RAFT text, precision increases to 1.0000, but recall drops to 0.6364. This means the model predicts fewer false positives, but it also misses more true disruptions. In contrast, the full model keeps a high recall while maintaining strong precision. Additional reliability checks are presented in Section 5.6.

5.4. Full Ablation Results

Table 7 reports the complete ablation results, including classification scores, relative changes, and learned modality gate weights.

5.5. Discussion

The results support the main claim of the paper. Maritime disruption detection requires grounded multimodal system design. Local physical signals provide the strongest evidence. Structured relations and retrieved documents improve interpretation and reduce unsupported LLM responses. A larger LLM alone does not provide the same balance. CauSAIL is useful because each prediction can be connected back to port-month evidence. AIS explains movement-level disruption. Sentinel-2 adds visual and port-water context. CT-SCKG links local behaviour to supply-chain entities and events. RAFT evidence helps the LLM distinguish core documents from tangent documents. The bias-aware and modality-aware components reduce overconfident reliance on one source. The confusion matrix provides that CauSAIL

Table 6
Performance metrics for ablation studies with different CauSAIL module combinations.

Modules					Performance							
SAT	AIS	KG	RAFT	LLM	Acc.	Prec.	Rec.	F1	AUC	AUPRC	MCC	ΔF1
✓	✓	✓	✓	✓	**0.9062**	0.8333	0.9091	**0.8696**	0.9610	0.9399	**0.7984**	**+0.0000**
✓	✓	×	×	✓	0.8438	0.6875	**1.0000**	0.8148	0.9524	0.9063	0.7237	-0.0548
✓	✓	×	×	×	0.8438	0.6875	**1.0000**	0.8148	0.9524	0.9063	0.7237	-0.0548
×	✓	×	×	×	0.8438	0.7143	0.9091	0.8000	0.8961	0.7751	0.6880	-0.0696
✓	✓	✓	×	✓	0.8750	**1.0000**	0.6364	0.7778	**0.9870**	**0.9751**	0.7311	-0.0918
✓	×	✓	✓	✓	0.6562	0.0000	0.0000	0.0000	0.6970	0.4965	0.0000	-0.8696
✓	×	×	×	×	0.6562	0.0000	0.0000	0.0000	0.5844	0.4621	0.0000	-0.8696
×	×	×	✓	✓	0.6562	0.0000	0.0000	0.0000	0.6017	0.4553	0.0000	-0.8696

Table 7
Full ablation results on the test split. Δ values are measured relative to the full CauSAIL model. Gate values show the average modality contribution learned by MA-LAFAM.

Ablation	Acc.	F1	Prec.	Rec.	AUC	AUPRC	MCC	ΔF1	ΔAcc.	ΔAUC	g_{SAT}	g_{AIS}	g_{KG}	g_{LLM}
FULL_MODEL	0.9062	0.8696	0.8333	0.9091	0.9610	0.9399	0.7984	+0.0000	+0.0000	+0.0000	0.2677	0.1566	0.2829	0.2927
NO_KG_AND_RAFT	0.8438	0.8148	0.6875	1.0000	0.9524	0.9063	0.7237	-0.0548	-0.0625	-0.0087	0.3719	0.2062	0.0000	0.4218
LOCAL_ONLY_AIS_SAT	0.8438	0.8148	0.6875	1.0000	0.9524	0.9063	0.7237	-0.0548	-0.0625	-0.0087	0.3719	0.2062	0.0000	0.4218
AIS_ONLY	0.8438	0.8000	0.7143	0.9091	0.8961	0.7751	0.6880	-0.0696	-0.0625	-0.0649	0.0000	0.2193	0.0000	0.7807
NO_RAFT_TEXT	0.8750	0.7778	1.0000	0.6364	0.9870	0.9751	0.7311	-0.0918	-0.0312	+0.0260	0.2738	0.1529	0.2696	0.3037
NO_AIS	0.6562	0.0000	0.0000	0.0000	0.6970	0.4965	0.0000	-0.8696	-0.2500	-0.2641	0.3258	0.0000	0.3245	0.3497
SAT_ONLY	0.6562	0.0000	0.0000	0.0000	0.5844	0.4621	0.0000	-0.8696	-0.2500	-0.3766	0.4865	0.0000	0.0000	0.5135
LLM_RAFT_ONLY	0.6562	0.0000	0.0000	0.0000	0.6017	0.4553	0.0000	-0.8696	-0.2500	-0.3593	0.0000	0.0000	0.0000	1.0000

Table 8
Bootstrap confidence intervals with $B = 1000$ resamples

Metric	Mean	Std.	95% CI Low	95% CI High
Accuracy	0.8429	0.0643	0.7188	0.9375
AUC	0.8809	0.0459	0.7857	0.9546
AUPRC	0.6842	0.1203	0.4000	0.8824
F1	0.8061	0.0894	0.5714	0.9375
MCC	0.7203	0.1000	0.5173	0.8819
Precision	0.6842	0.1203	0.4000	0.8824
Recall	1.0000	0.0000	1.0000	1.0000

misses one disrupted port-month and creates two false positives. For an industrial monitoring team, this system can serve as a screening layer. Analysts can review the small number of flagged cases with attached KG and RAFT evidence. We provide an additional qualitative analysis in Section 5.7 to illustrate how correct and incorrect predictions are distributed within the learned test-space structure.

5.6. Additional Reliability Simulations

We run additional reliability checks with a threshold fixed at 0.45. We examine metric uncertainty, probability calibration, seed sensitivity, port-identity shortcuts, time-leakage risk, and sensitivity to tangent RAFT documents.
Table 8 shows that the test split creates visible uncertainty around several metrics. Recall remains stable because the threshold model detects all positive cases in this robustness run. Precision, AUPRC, F1, and MCC show wider intervals, which means future evaluation should include larger and more diverse port-month samples.
Table 9 reports a Brier score of 0.1781. True positives and false positives have the same average predicted probability of 0.55, which suggests that the model separates many positive and negative cases but still assigns similar confidence to some incorrect positive predictions.

Table 9
Brier score calibration summary. The Brier score is 0.1781.

Outcome	Count	Average predicted probability
TP	11	0.55
TN	16	0.35
FP	5	0.55
FN	0	–

Table 10
Seed sensitivity results

Seed	Threshold	Accuracy	Precision	Recall	F1	AUC	AUPRC	MCC
1000	0.45	0.8438	0.6875	1.0000	0.8148	0.9394	0.8907	0.7237
1001	0.60	0.7500	0.5000	0.5000	0.5000	0.6667	0.4103	0.3333
Mean	–	0.7969	0.5938	0.7500	0.6574	0.8030	0.6505	0.5285
Std.	–	0.0663	0.1326	0.3536	0.2226	0.1928	0.3397	0.2761

Table 11
Port-identity test

Setting	Accuracy	Precision	Recall	F1	AUC
Original	0.8438	0.6875	1.0000	0.8148	0.8810
Port ID removed	0.8438	0.6875	1.0000	0.8148	0.8810
Port ID shuffled	0.8438	0.6875	1.0000	0.8148	0.8420

Table 12
Time-leakage audit

Setting	Accuracy	Precision	Recall	F1	AUC	AUPRC	MCC	ΔF1	ΔAUC
Original	0.8438	0.6875	1.0000	0.8148	0.8810	0.6875	0.7237	0.0000	0.0000
Lagged pressure	0.9375	0.9091	0.9091	0.9091	0.9870	0.9778	0.8615	+0.0943	+0.1061
No current pressure	0.8438	0.6875	1.0000	0.8148	0.8268	0.6100	0.7237	0.0000	-0.0541

Table 10 shows sensitivity across two random seeds. The first seed gives stable disruption detection, while the second seed reduces recall and F1.
Table 11 reports no prediction flips after removing or shuffling port identity. This suggests that the model does not rely only on port labels for its thresholded predictions. The AUC decreases under the shuffled setting, so port identity may still affect ranking behaviour.
Table 12 shows that the model does not depend only on current-month pressure. The lagged-pressure setting improves several metrics in this run, while removing current pressure leaves F1 unchanged but lowers AUC. This suggests that temporal pressure variables affect ranking more than thresholded classification.
Table 13 shows that tangent-document changes do not flip predictions at the selected threshold. F1 and MCC stay fixed, while AUC and AUPRC vary. This means RAFT noise affects ranking confidence more than the final thresholded decision in this test.

5.7. Topological Visualization of Test Predictions

We include a topological visualization to inspect how the test samples are arranged in the learned representation space. Figure 3 projects the test port-month representations into a two-dimensional space. Each point represents one test sample. Edges connect nearby samples in the projected space, showing local neighborhood structure. The colour of

Table 13
Tangent-document stress test

Tangent/core ratio	FP rate	F1	AUC	AUPRC	MCC
0	0.2381	0.8148	0.8810	0.6875	0.7237
1	0.2381	0.8148	0.8052	0.5991	0.7237
2	0.2381	0.8148	0.8874	0.6997	0.7237
3	0.2381	0.8148	0.8766	0.6799	0.7237

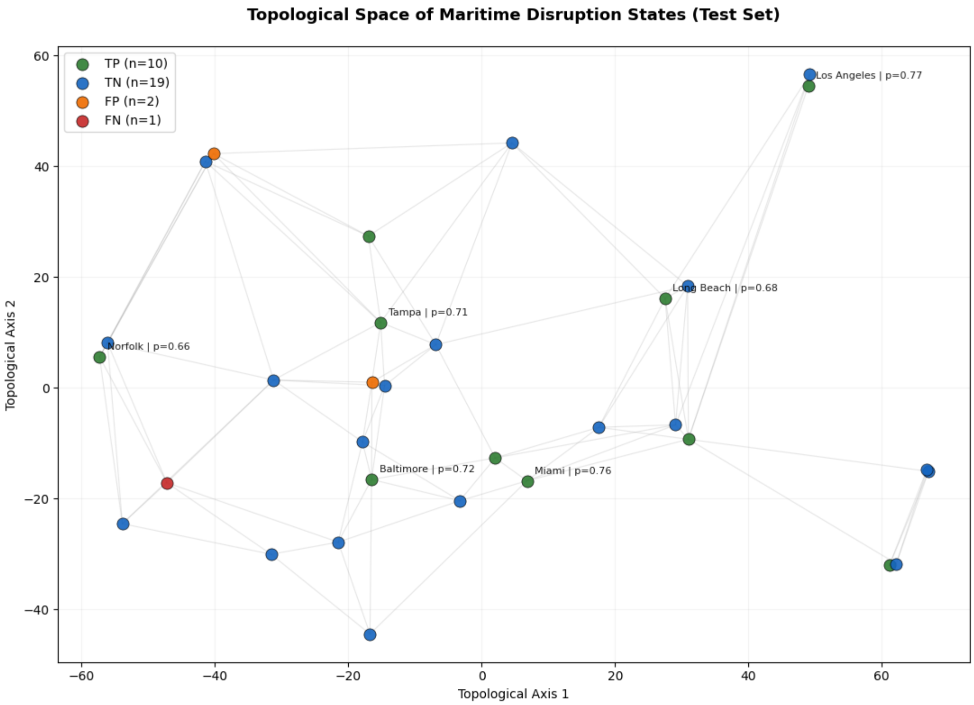


Figure 3: Topological space of maritime disruption states on the selected-local test split. Green points denote true positives, blue points denote true negatives, orange points denote false positives, and red points denote false negatives.

each point indicates the prediction outcome: true positive (TP), true negative (TN), false positive (FP), or false negative (FN). This figure is qualitative. We use it to understand the error pattern, not as a replacement for the quantitative scores reported in Table 4.

The visualization shows three core findings. First, most test samples are correctly classified. The plot contains 10 true positives and 19 true negatives, which matches the confusion matrix result. This indicates that the learned representation separates many normal and disrupted port-months into stable local neighborhoods.

Second, the disrupted cases with high predicted probabilities appear as clear positive-risk states. Several labelled true positives, such as Los Angeles, Miami, Baltimore, Tampa, Long Beach, and Norfolk, receive high disruption probabilities. These cases suggest that CauSAIL learns consistent risk patterns when vessel movement, satellite context, graph evidence, and RAFT evidence support the same decision.

Third, the error cases appear in boundary regions. The two false positives are close to nearby normal or mixed neighbourhoods, which suggests that the model identifies them as risk-like even though the ground truth label is normal. The single false negative appears away from the strongest positive-risk region. This pattern is important for deployment because it shows where the model may need additional evidence, better temporal context, or analyst review.

Overall, Figure 3 supports the main result analysis. CauSAIL not only produce strong aggregate scores. It also forms a representation space where most correct predictions occupy coherent local neighbourhoods, while errors are sparse and interpretable. This motivates the use of modality-aware fusion and reliability checks for port-month maritime disruption detection.

6. Limitations

CauSAIL still has several limitations for future work.

- Real-time deployment should be evaluated
- Computational resources should be expanded
- Short disruption windows should be tested
- Causal effects should be further tested

7. Conclusion

Our experiments show that maritime disruption detection improves when vessel movement, satellite signals, structured supply-chain relations, and retrieved textual evidence are integrated into a single port-month system. CauSAIL aligns seven data sources, grounds decisions with CT-SCKG and RAFT evidence, and fuses AIS, Sentinel-2, KG, and LLM states through MA-LAFAM. Results show stronger decision balance than LLM backbones alone, while ablations confirm that AIS is the strongest positive disruption signal and that graph grounding, retrieval control, and bias-aware fusion reduce unsupported decisions. Future work will scale CauSAIL to broader ports, real-time evidence streams, and analyst-in-the-loop validation.

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